SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

CO1-AT2-Constraint-Based Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 20%

QUESTIONS:3 **Total Marks:30**

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift
   ii. D2 must work before D3 (Morning < Afternoon < Night)
   iii. Only one doctor per shift
   iv. D3 cannot work Morning
   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.
   The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

   Constraints:

   - The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:
a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

I Ganji Madhan Kumar(192472328) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **G.Madhan Kumar**

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5

Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

Q1. Hospital Doctor Assignment using Backtracking

Doctors: D1,D2,D3; Shifts: Morning, Afternoon, Night.

Constraints:
1. D1 cannot work Night.
2. D2 must work before D3.
3. One doctor per shift.
4. D3 cannot work Morning.
5. Every doctor gets exactly one shift.

Backtracking Steps:
Step 1: Assign D1→Morning.
Step 2: Assign D2→Afternoon.
Step 3: Remaining shift is Night, assign D3→Night.

Constraint Check:
• D1 not Night
• D2 before D3
• D3 not Morning
• One doctor per shift
• Every doctor assigned once

Final Schedule:
Morning–D1
Afternoon–D2
Night–D3

Q2. Robot Navigation using BFS

Note: The question paper does not specify the 5×5 obstacle layout. The following uses an assumed grid for demonstration.

Grid:

```
Grid:
S . . X .
. X . X .
. X . . .
. . X X .
X . . . G

Shortest Path:
[(0, 0), (1, 0), (2, 0), (3, 0), (3, 1), (4, 1), (4, 2), (4, 3), (4, 4)]

Total Cost: 8

Path on Grid:
S . . X .
* X . X .
* X . . .
* * X X .
X * * * G
```

BFS expands nodes level by level using a queue.

Expansion:
1. Start S.
2. Visit (1,2), (2,1).
3. Visit (1,3).
4. Visit (3,1).
5. Visit (2,3).
6. Continue until Goal is reached.

One optimal path:
S → (2,1) → (3,1) → (3,2) → (3,3) → (3,4) → (3,5) → (4,5) → G

Cost = 8 moves.

Q3. Autonomous Rescue Robot in a Disaster-Affected Building.

a) Analysis of the Constraints and Their Effect on Decision-Making

Constraint	**Effect on Robot's Decision-Making**
Robot can move up, down, left, right	Limits movement to four directions, simplifying possible actions.
Blocked cells (obstacles)	Robot must detect and avoid obstacles while searching for a valid path.
Limited sensing (adjacent cells only)	Robot does not know the complete environment initially and must explore gradually.
Movement cost = 1	Every step contributes to the total path cost.
Risky zones have additional cost of 2	Robot should avoid risky areas whenever possible, even if the path is slightly longer.
Minimize total cost	The robot must choose the least-cost path instead of simply the shortest path.
Dynamic knowledge update	As new information is discovered, the robot updates its map and replans its path if necessary.

Decision-Making Summary:

- Avoid blocked cells.
- Prefer safe routes over risky ones.
- Continuously update knowledge from nearby observations.
- Select the path with the minimum total cost.

b) Solution Approach

Search Strategy: Uniform Cost Search (UCS) with Online Search

Step 1: Initialize

- Start at S.
- Goal is G.
- Initialize total path cost = 0.
- Robot knows only adjacent cells.

Step 2: Sense Environment

- Observe neighboring rooms.
- Identify:
    - Safe cells
    - Risky cells
    - Obstacles

Step 3: Expand Lowest-Cost Node

- Store possible moves in a priority queue.
- Always choose the node with the smallest cumulative cost.

Step 4: Update Knowledge

Whenever new rooms become visible:

- Add newly discovered cells.
- Remove blocked paths.
- Update movement costs.

Step 5: Replan if Necessary

If an obstacle blocks the planned route:

- Recalculate the lowest-cost path.
- Continue toward the goal.

Step 6: Reach Goal

Stop when the survivor (**G**) is reached with minimum total cost.

Algorithm

Start at S
Initialize Priority Queue
While Goal not reached:
Observe adjacent cells
Update map
Expand lowest-cost node
Ignore blocked cells
Add movement cost
Add extra cost for risky cells
Choose lowest cumulative cost
Return path to Goal

c) Application of AI Concepts

1. Intelligent Agent

The rescue robot is an **Intelligent Agent** because it:

- Perceives the environment through sensors.
- Makes decisions using observations.
- Acts through movement.
- Learns new information while exploring.

PEAS Representation

Component	**Description**
Performance Measure	Reach survivor safely with minimum cost
Environment	Disaster-affected building
Actuators	Wheels/Motors for movement
Sensors	Cameras, proximity sensors, obstacle detectors

2. Rationality: The robot is rational because it always chooses the action that minimizes the expected total cost based on currently available information.

Example:
Path A:5 safe moves
Cost = 5
Path B:3 moves through risky zone
Cost = 3 + 2 = 5
If another safe path costs only **4**, the robot selects that path since it has the lowest total cost.

3. Environment Type

Property	**Type**
Observability	Partially Observable
Agents	Single Agent
Determinism	Deterministic
Dynamics	Dynamic (knowledge changes during exploration)
State Space	Discrete
Decision Type	Sequential

The environment is **partially observable** because the robot can only sense adjacent rooms.

d) Justification of the Chosen Approach

The Uniform Cost Search (UCS) combined with an Online Search Agent is the most suitable approach because:

- Finds the minimum-cost path, considering both movement and risk costs.
- Handles different action costs effectively.
- Supports dynamic replanning when new obstacles are discovered.
- Works well in partially observable environments by updating knowledge during exploration.
- Ensures safe navigation while minimizing total travel cost.